NATURAL LANGUAGE PATH FINDING

WALKABLE PATHS, FROM EVERYDAY INSTRUCTIONS

SHASHI BHIDODIYA KRISHNA KUMAYU

SUPERVISOR: PROF. INDRANIL SAHA
DEPT. OF CSE, IIT KANPUR

UNDERGRADUATE PROJECT

SECTION 1

The Problem

WHAT DO PEOPLE ACTUALLY SAY?

"Go to the lab, don't go through the workshop,
stop at the store room first, and stay in the corridor if you can."

Goal
go to the lab

Never
not the workshop

In order
store room first

If you can
prefer the corridor

A normal path planner (A*) only understands **one goal cell**.

TWO EXISTING WAYS, AND BOTH HAVE A PROBLEM

1. Ask the LLM for the path

- The LLM reads the map and writes the cells
- Understands any sentence

But: nothing stops it from walking **through walls**

2. Translate to a logic formula

- LLM writes a formula (LTL), a planner follows it
- Correct *if* the formula is correct

But: formulas are **easy to get wrong**, hard to read, and cannot say “if you can”

We want: **any sentence** in, a **safe and correct path** out.

SECTION 2

Our Idea

THE LLM DECIDES *WHAT*, A* DECIDES *HOW*

The LLM only fills a small form.
It **never sees the map** and never writes a cell.

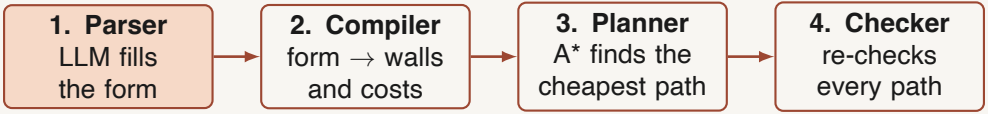
Intent form (from the sentence)

goal	lab
forbidden	workshop
stops	store room
prefer	corridor
avoid	—

Why a form?

- Easy for the LLM to fill
- Easy for a person to read and check
- Can say “if you can”
- Wrong names are caught and sent back (at most 3 tries)

THE PIPELINE



Forbidden room
→ becomes a wall

Prefer = cheaper (0.5)
Avoid = costlier (3)

Stops in order
→ planner counts the stops done so far

WHAT WE CAN GUARANTEE

PROVED ON PAPER, AND TESTED ON 1000S OF RANDOM MAPS

1. The path is **always walkable**, even if the LLM misunderstood the sentence.
2. It follows every rule in the form (never, in order, goal).
3. If we say `NO_PATH`, then **no path exists**.
4. It is the **cheapest** such path.
5. “If you can” never causes a huge detour (bounded).

What is *not* guaranteed

That the LLM reads the sentence correctly, so we **measure** it.

SECTION 3

Experiments

SETUP

Benchmark

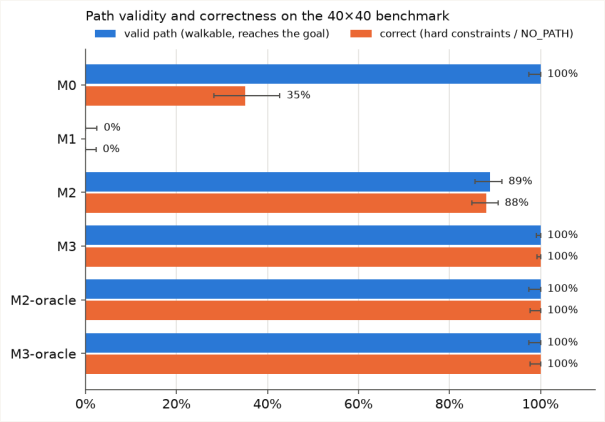
- 20 generated floor plans (40 × 40 grid)
- 165 instructions, each in **3 wordings** = 495
- Types: plain, never, stop, in order, “if you can”, impossible
- 30 written by hand, deliberately messy

Methods compared

- **M0** plain A* (ignores the rules)
- **M1** LLM writes the path
- **M2** LLM writes a logic formula
- **M3** ours (intent form)

Same LLM for all: Google Gemini

MAIN RESULT

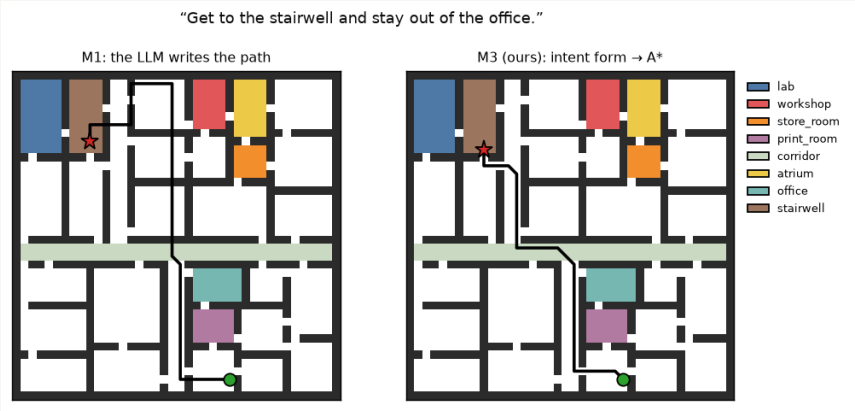


Method	Correct
M0 plain A*	35%
M1 LLM writes path	0%
M2 logic formula	88%
M3 ours	**100%**

On “in order” instructions the formula method is right only **61%** of the time.

Oracle = the planner is given the correct form / formula directly (no LLM).

LLM WRITING THE PATH: THROUGH THE WALLS



160 of 161 LLM-written paths were **not walkable**. Ours: all walkable.

WHY DOES THE FORMULA METHOD LOSE?

	Form	Formula
LLM got it right	**100%**	56%
Same for all 3 wordings	**100%**	30%

In short

- Both planners are perfect when given the right input
- So the loss is in the **LLM's translation**
- Filling a form is much easier than writing logic

“IF YOU CAN”: ONLY THE FORM CAN TRADE OFF



Ignore it (left) · make it a strict rule, long detour (middle) · **ours: balanced** (right)
Ours spends about **twice as much** of the path in the preferred corridor, for paths only **9%** longer on average.

SECTION 4

Next Steps

LIMITATIONS

- The LLM's reading is not guaranteed, only measured
- Benchmark is generated: short sentences, known room names
- One LLM (Gemini Flash-Lite) tested
- Not yet tested with a real robot
- M1 and M2 are **our own versions** of the two approaches, not the published systems

FUTURE WORK

Compare with published LLM planners

- **LLM-A***: LLM suggests waypoints, A* joins them
- **LLM+P**: LLM writes a problem for a classical planner
- Published NL-to-logic systems

Make it more realistic

- Real building maps and real people's instructions
- Testing with the ROS/Gazebo simulator
- Then test on a real mobile robot

Thank you!